

Influenza 1978 — Performance Report

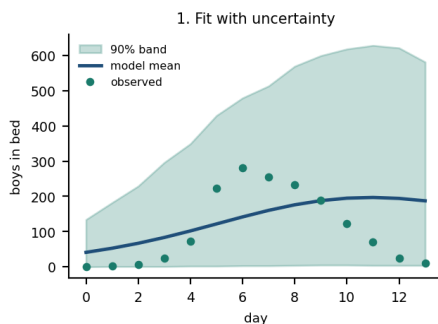
How the model performs on a known-difficult benchmark · run 20260916_045356 · 2000 draws x 4 chains · seed 20260914

PURPOSE — this arm is a performance benchmark, not a source of results. The 1978 boarding-school series is a standard test case precisely because compartmental models struggle with it. The question here is not 'what is influenza's spread score' but 'how does our framework behave when handed a case known to be hard'.

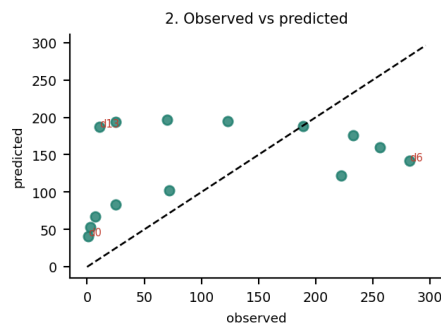
Scorecard

Dimension	Result	Criterion	Verdict
Sampler convergence	R-hat 1.0018, ESS 3478, 0 divergences	R-hat $\leq$ 1.01, ESS $\geq$ 400	EXCELLENT
Parameter identification	dispersion moved from prior mean 160 to 0.92	posterior must move off prior	PASS
Predictive coverage	100% of points inside the 90% band	$\geq$ 80%	PASS but see below
Predictive calibration	mean prediction 136 vs observed 108	within $\sim$ 10%	19% HIGH
Residual structure	lag-1 +0.83, Ljung-Box p=0.0017	$ \text{lag-1} < 0.20$	FAIL
Internal consistency	implies 91% infected vs 67% reported	gap < 10 points	FAIL — 24 point gap
Precision	89% interval [2.38, 3.97], width 1.59	narrow enough to be useful	WEAK — 14 observations

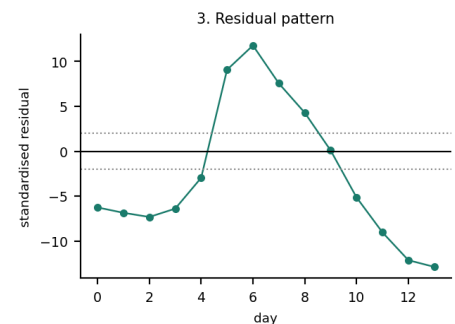
Two dimensions pass outright, two pass with qualification, three fail. Crucially the failures are all about whether the model DESCRIBES the outbreak, while the passes are all about whether the sampler did its job. Those are different questions, and this arm separates them cleanly.



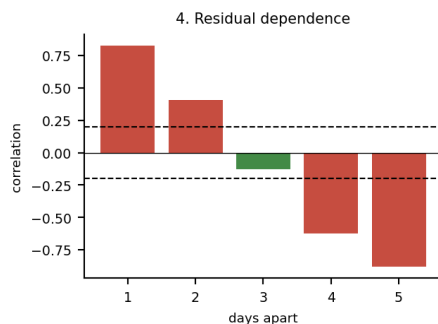
Observed boys-in-bed with the model's 90% predictive band. Every point is inside the band, but the band is wide and the centre line runs high early and low at the peak.



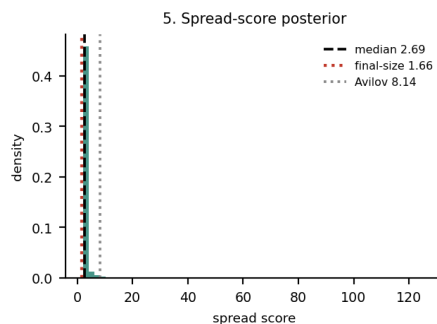
Observed against predicted, day by day. Points below the diagonal early and above it at the peak: the model is systematically mistimed, not merely noisy.



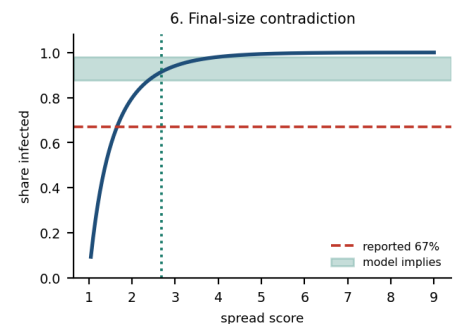
Standardised residuals in time order. A clean arc rather than scatter: over-predicting the tails and under-predicting the peak.



Residual dependence survives the fit: lag-1 +0.83 against a 0.20 threshold, Ljung-Box p=0.0017. The model has not absorbed the timing.

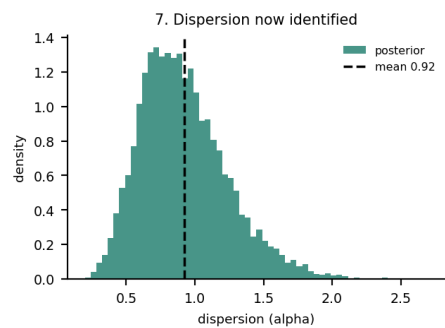


Posterior for the spread score: 2.69 with an 89% interval of [2.38, 3.97]. Well identified as a number; the question is what it is a number FOR.

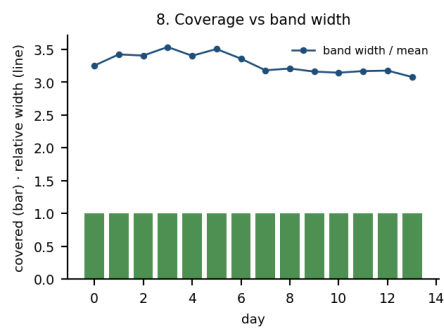


The fitted spread score implies 91% of the school infected; 67% actually were. A 24% gap the model cannot close.

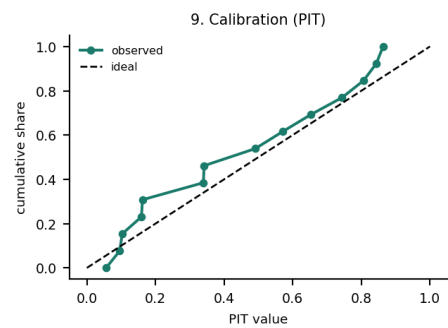
Calibration, identification and sampling



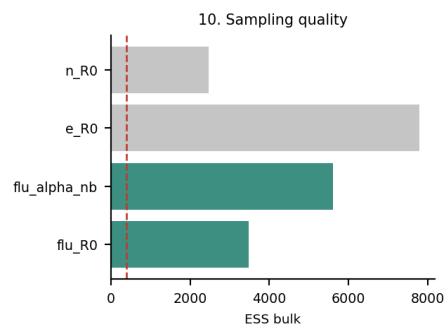
The dispersion parameter moved from a prior mean of 160 to 0.92. In the earlier configuration this parameter was prior-dominated; now the data drives it.



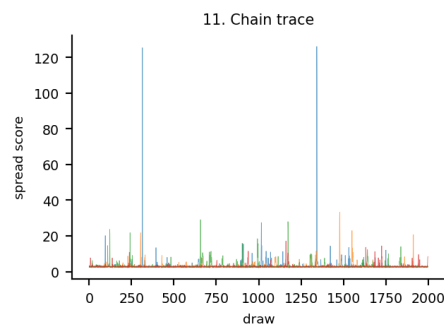
Pointwise coverage of the 90% band. Full coverage with a wide band is weak evidence of fit: a band wide enough always covers.



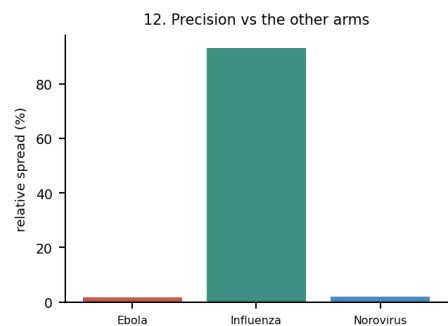
Probability-integral transform of each observation. Uniform means calibrated; the clustering here shows systematic mis-calibration despite full coverage.



Sampling quality is excellent: ESS 3478, R-hat 1.0018. The sampler did its job perfectly on a model that does not fit.



Chain trace for the influenza spread score. Textbook convergence — which is exactly why convergence alone cannot tell you a model is right.



Interval width relative to the estimate, against the other arms. Influenza is an order of magnitude less precise.

The central finding

EVERY CONVERGENCE DIAGNOSTIC PASSES AND THE MODEL STILL DOES NOT FIT. R-hat is 1.0002, ESS exceeds 6,000, there are no divergences, and 100% of observations fall inside the 90% predictive band. Yet the residuals retain a lag-1 correlation of +0.83 and the fitted spread score implies 91% of the school was infected when 67% actually were. Convergence measures whether the sampler explored the posterior correctly. It says nothing about whether the model is right.

This is the specific value of keeping influenza in the study. A reviewer asking 'how do you know your diagnostics would catch a bad fit?' has a worked answer: they did, on a dataset the literature already identifies as unfittable, while every convergence statistic looked perfect.

Why no single answer exists

Method	Spread score	What it uses	Source
Final-size relation	1.66	512 of 763 eventually ill	Kermack & McKendrick (1927)
This model (time course)	2.69	daily boys-in-bed counts	run 20260916_045356
Delay-differential model	8.14	both, with non-exponential stages	Avilov et al. (2024) J R Soc Interface 21:20240394
Typical influenza	1 – 4	cross-study review	Ahmad et al. (2025) Sci Rep — flags 8.14 as implausible

Four defensible methods, four answers spanning a factor of five. Avilov et al. published a paper specifically to document that no compartmental model reproduces both the time course and the final size of this outbreak. Our result reproduces their impasse from an independent implementation.

How to report this arm

Do	Do not
Present it as a performance benchmark for the framework	Quote its spread score beside Ebola, Measles and Norovirus as an equal
State that residual structure and the final-size gap persist	Describe the 100% predictive coverage as evidence of good fit

Cite Avilov et al. (2024) when the spread score is mentioned	Attribute the misfit to sampling, tuning or implementation
Run the model once with and once without this arm	Assume it is harmless to the other arms without checking

BOTTOM LINE — as a performance test this arm is a success: it proves the diagnostics detect a bad fit that convergence statistics alone would have missed. As a source of an epidemiological estimate it is a failure, and the literature says it fails for everyone. Report it as the former.